Question	Answer	Marks	Partial Marks
4(b)(i)	$2(x + 12y)(x – 12y)$ final answer	**3**	**B2** for $(2x + 24y)(x – 12y)$ or $(2x – 24y)(x + 12y)$ or for $2(x + 12y)(x – 12y)$ seen OR **M2** for $k(x + 12y)(x – 12y)$ or **M1** for $2(x^2 – 144y^2)$
4(b)(ii)	$(5x – 8)(x + 5)$ final answer	**2**	**M1** for $5x(x + 5) – 8(x + 5)$ or $x(5x – 8)+ 5(5x – 8)$ or for $(5x + a)(x + b)$ where $ab = – 40$ or $a + 5b = 17$
4(c)	$4x^2 – 17x + 9$ [= 0] oe	**B1**	
	$\frac{[--]17 \pm \sqrt{([-]17)^2 - 4(4)(9)}}{2\times 4}$	**B2**	**FT** *their* 3 term quadratic **B1FT** for $\sqrt{([-]17)^2 - 4(4)(9)}$ or better or $\left(x - \frac{17}{8}\right)^2$ oe or $\sqrt{\frac{([-]17)^2 - 4(4)(9)}{4}}$ or better and **B1FT** for $\frac{[--]17 + \sqrt{q}}{2(4)}$ or $\frac{[--]17 - \sqrt{q}}{2(4)}$ or better or $\frac{17}{8} + \sqrt{\frac{145}{64}}$ oe or $\frac{17}{8} - \sqrt{\frac{145}{64}}$ oe or $\frac{\frac{[--]17}{2} + \sqrt{q}}{4}$ or $\frac{\frac{[--]17}{2} - \sqrt{q}}{4}$
	0.62 and 3.63 cao	**B2**	**B1** for each **SC1** for 0.6[0] or 0.619 to 0.620 **and** 3.6[0] or 3.6301 to 3.6302 or 0.62 and 3.63 seen in working or –0.62 and–3.63 as final answers
5(a)(i)	62 and Angle at centre is twice angle at circumference oe	**2**	**B1** for either
5(a)(ii)	117 and Isosceles [triangle] and Opposite angles in a cyclic quadrilateral are supplementary	**4**	**B2** for 117 or **B1** for [angle *OCD* =] 28 **B1dep** for isosceles [triangle] and **B1** for opposite angles in a cyclic quadrilateral are supplementary

Question	Answer	Marks	Partial Marks
5(b)	24.9 or 24.94 to 24.95	**5**	**B1** for angle $PQS = 42$ **M2** for $QS = 5.9 \div \cos 42$ oe or **M1** for $\cos 42 = \frac{5.9}{QS}$ oe **M1**dep for *their* $SQ \times \pi$ oe
6(a)(i)	9.5, 4.8 and 8.5	**3**	**B1** for each
6(a)(ii)	correct curve	**5**	**B4** for correct curve, but branches joined or touching y axis or **B3FT** for 9 or 10 correct plots or **B2FT** for 7 or 8 correct plots or **B1FT** for 5 or 6 correct plots AND **B1 indep** two separate branches not touching or cutting y-axis
6(b)	$y = \frac{24}{5} - 2x$ ruled and – 0.4 to – 0.2 and 1.45 to 1.7	**4**	**B2** for correct ruled line crossing curve twice or **B1** for correct freehand or for short ruled line or for line with negative gradient through (0, 4.8) or for line with gradient – 2 **B1** for each value
6(c)	[a =] 10 [b =] 20 [c =] – 48	**4**	**B3** for $10x^3 - 15 = 48x - 20x^2$ oe or better or **B2** for 2 correct values or **B1** for 1 correct value or for $5x^2 - \frac{15}{2x} = 24 - 10x$ or better or for $2x^3 - 3 = \frac{48}{5}x - 4x^2$ or better or for $x^3 - \frac{3}{2} = \frac{24}{5}x - 2x^2$ After 0 scored SC1 for correct elimination of a denominator of 5, x or $2x$ from a four term expression.
7(a)(i)(a)	Shape at (–2, 1) (–4, 1) (–4, 7) (0, 7)	**2**	**B1** for 3 correct points or for enlargement SF2 from any centre
7(a)(i)(b)	Shape at (2, –2) (2, –3) (5, –1) (5, –3)	**3**	**B2** for correct orientation but wrong position or for 3 correct points or **B1** for $y = x - 1$ drawn